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Quadrature schemes for integrals of density functional theory

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The evaluation of integrals which arise in density functional theory, as applied to molecules, as discussed. Becke's scheme for reducing them to a sum of integrals over atom based polyhedra is used. Within each of these regions, quadratures for the spherical polar coordinates are examined; in particular we compare a Euler-Maclaurin based scheme with Gauss schemes. Upon specific investigations we find that the Euler-Maclaurin scheme is favoured for radial quadrature and Gauss-Legendre quadrature is preferred for theta. We investigate the number of quadrature points required for a given accuracy, and we demonstrate our favoured approach by calculations on a variety of molecules.

1. Introduction

There are a growing number of computational chemists today who are using the ideas of density functional theory to calculate molecular properties. At the simplest level the Hohenberg-Kohn [1] and the Kohn-Sham [2] theories state that given an appropriate density $\rho(\mathbf{r})$, the exchange, correlation and kinetic energy correction contributions to the energy are given by

$$I = \int F(\rho, \nabla\rho, \dots) d\mathbf{r} \quad (1)$$

where F is a function of the density $\rho(\mathbf{r})$. Most certainly F is of such a complicated form that it is not possible to calculate the integral analytically. It is known for example that a good first approximation to the exchange contribution is proportional to $\rho^{4/3}$, and more accurate forms are more complicated. As far as is known F is differentiable almost everywhere and thus F contains no singularities. The behaviour of F is governed by the behaviour of $\rho(\mathbf{r})$, which for nearly all molecules (a well-known exception being Li_2 [3]), has cusp shaped maxima at the nuclei and decreases to zero as $r \rightarrow \infty$.

Our initial philosophy towards density functional theory (DFT) is as follows. We, being quantum chemists, have a good self consistent field (SCF) package which uses standard procedures (Rys quadrature or recurrence relations) for the evaluation of the two electron Gaussian basis function integrals. For larger molecules the direct methodology is used which means that the two electron integrals are calculated as required, but not stored. To include electron correlation effects, our simplest procedure is the Møller-Plesset second-order (MP2) approach. Whilst for small molecules there are more sophisticated methods (e.g. coupled cluster), these methods are today too expensive for larger molecules. MP2 can be used for some larger molecules using direct algorithms (say up to 500 basis functions), and SCF can be applied to even larger systems. We wish to enquire whether DFT is a competitor to MP2, and indeed if it is better than SCF. There are now a substantial number of calculations

which suggest that the answers to these questions is 'yes'. We wish to examine this question making a minimal number of approximations. To do this initially, we are taking functionals F from the literature. We then examine two approaches: (i) we use the density from a converged SCF calculation, and then simply replace the exchange part of the total energy by the integral of an exchange functional, and also add on a correlation energy determined as an integral of a correlation functional; and (ii) we replace the exchange operator in the SCF equations by the Euler–Lagrange potentials derived exactly from the above exchange and correlation functionals, as is prescribed in the Kohn–Sham method. In both cases, the Coulomb part of the problem is evaluated exactly as in our standard programs. In order to examine these two (standard) approaches we need to be able to evaluate the integrals (1) to any desired degree of accuracy. The purpose of this paper is to describe the quadrature scheme which we use, one which is capable of producing energies for the above two schemes to a numerical accuracy of 10^{-10} hartree, if it is desired. We do desire this accuracy if we are going to examine the above two schemes in the way described. We are not aware that this has been done before for the more sophisticated functionals in the literature. We are convinced of the importance of such numerical checks, and also the ability of all workers in this field to reproduce one another's work, which has been the hallmark of quantum chemistry.

There are a number of commercial DFT codes available, which are of increasing use to the community as functionals are improved. A strict comparison of these codes with our work is not appropriate at this stage because we are trying to solve these problems with no approximations. The success of the commercial DFT codes depends upon further enhancements such as fitting the functionals concerned to Gaussians. Our work is of a more fundamental nature at this stage of development.

The integral, I , has normally been calculated by numerical quadrature and the importance of doing this efficiently can not be over-estimated since it can be the most time consuming step in molecular applications. We recognize that many workers have considered and used molecular grid quadratures over the years. In the early days Boys and Rajagopal [4] gave a scheme which apportioned molecular space into atomic parts. Boys and Handy used this idea in their molecular trans-correlated studies [5]. The latter authors used a three-dimensional integration over the unit cube which integrated Fourier functions exactly. Such grid points arose from solving a set of diophantine equations, a method originally due to Haselgrove [6]. Conroy [7] also worked in this area. More recently Friesner's success in his pseudo-spectral method depended upon the introduction of molecular grids which were again atom based [8]. He used radial Chebyshev points and small angular grids. Baerends and coworkers (see for example reference [9]) have used the method proposed by Ellis and Painter [10] which is based on diophantine theory [6, 7]. Boerrigter *et al.* [11] later found that Gauss–Legendre in the r , θ and ϕ directions gave superior convergence as the number of quadrature points increased, they also showed how symmetry could be used with their method to restrict the number of points. Becke [12] gave a straightforward partition of molecular space into overlapping atomic fragments. He recommended the use of Gauss–Chebyshev quadrature in the radial direction and Stroud–Lebedev quadrature [13, 14] for the angular integration. Andzelm and Wimmer use this approach in their Dgauss program (see for example reference [15]) and also offer some useful tips on reducing the number of quadrature points in actual calculations. Very recently te Velde and Baerends [16] have produced a complete paper on this topic, which has some overlap with the

contents of this paper. We shall point out similarities and differences as we proceed (we shall refer to their work as tVB).

In this paper we examine different quadrature schemes and our aim is to make a contribution towards answering the question of which of the many possible methods require the least number of points in general applications. Taken with other papers on this topic, we shall conclude that the evaluation of I to a sufficient accuracy is not a serious problem. We compare the different approaches for some representative systems and then test what we consider to be the best for a large number of molecules. Our test cases (and our method) should easily be reproducible by other workers and we hope that this will allow other approaches to be tested on an equal footing so that some general conclusions can be reached.

In particular we shall discuss and use theory from two previous papers. The first, by Becke [12], describes a simple scheme for the decomposition of the integral I over three-dimensional space into a sum of integrals each of which is based on a nucleus of the molecule. The use of this decomposition is essential to our scheme. Following Becke, we then use spherical polar coordinates for the evaluation of each of these atomic-like integrals. For the quadrature scheme for each of the polar coordinates we shall use numerical grids which are based upon either the Euler–Maclaurin formula or Gauss quadrature. We shall see that the attraction of the Euler–Maclaurin scheme is that it appears to be successful for almost any form of function, in contrast to Gauss-like formulae which are ideal for the case when the precise form of the integrand is known; in some cases Gauss quadrature schemes can be used to exactly evaluate the integral and save complicated algebra. To quantum chemists the best known example is the Rys procedure [17] for the evaluation of four-centre two-electron integrals over Gaussian basis functions. It does not follow that Gauss-based formulae are best when the integral cannot be analytically evaluated. Indeed we do not expect that there is a unique answer to this problem of obtaining a best quadrature scheme, and our proposed scheme will be one of many.

The second paper, by Handy and Boys [18], describes the essential ideas behind a quadrature scheme based on the Euler–Maclaurin formula. The key is that this formula gives the error in a simple quadrature scheme, and it is possible to eliminate successive error terms by appropriate choice of the Jacobian in a change of variable. In [18] several examples were given to demonstrate the practicality of the scheme. In section 2 the theory of this quadrature scheme is given, with its application to integrations over the three spherical polar coordinates. In section 3 the ideas of Becke are incorporated to define procedures for the evaluation of general integrals of the form (1). In section 4 the forms of function which are integrated are given and section 5 examines the relation between accuracy and the number of quadrature points. In section 6 we show how we have used symmetry to reduce the number of required quadrature points. The reliability of our scheme is demonstrated by explicit calculations on many molecules in section 7.

2. The Euler–Maclaurin and Gauss quadrature schemes

2.1. The Euler–Maclaurin formula

The Euler–Maclaurin formula, in the context that we wish to use it, concerns the evaluation of an integral in one dimension using equally spaced points and equal weights. We shall consider $[0, 1]$ divided into n equal parts, with quadrature points at

i/n ($i = 0, 1, 2, \dots, n$). We also consider a change of variable from x to q and so the integral we are considering is

$$\int_0^1 F(x) dx = \int_0^1 F(x(q)) \frac{dx}{dq} dq \equiv \int_0^1 G(q) dq. \quad (2)$$

The Euler–Maclaurin formula [19] for the evaluation of the integral (2) is

$$\begin{aligned} \int_0^1 G(q) dq &= \frac{1}{n} \left[\sum_{i=1}^{n-1} G\left(\frac{i}{n}\right) + \frac{1}{2}[G(0) + G(1)] \right] \\ &\quad - \sum_{k=1}^m \frac{B_{2k}}{(2k)!} \left(\frac{1}{n}\right)^{2k} [G^{(2k-1)}(1) - G^{(2k-1)}(0)] \\ &\quad - \frac{B_{2m+2}}{n^{2m+1}(2m+2)!} G^{(2m+2)}(\xi) \end{aligned} \quad (3)$$

where $0 < \xi < 1$. B_{2k} are Bernoulli numbers which are 1, $1/6$, $-1/30$, $1/42$, $-1/30$, $5/66$, $-691/2730$, $7/6$, $-3617/510$, $43867/798$, $-174611/330$ for $k = 0, \dots, 10$. The Bernoulli numbers grow very rapidly and for large even k , $B_k \simeq 2(-1)^{k-1}(2k)!(2\pi)^{-2k}$. Thus the series in (3) is only useful for low values of m . The Euler–Maclaurin formula becomes

$$\begin{aligned} \int_0^1 G(q) dq &= \frac{1}{n} \left[\sum_{i=1}^{n-1} G\left(\frac{i}{n}\right) + \frac{1}{2}[G(0) + G(1)] \right] - \frac{1}{12n^2} [G^{(1)}(1) - G^{(1)}(0)] \\ &\quad + \frac{1}{720n^4} [G^{(3)}(1) - G^{(3)}(0)] - \frac{1}{30240n^6} [G^{(5)}(1) - G^{(5)}(0)] \\ &\quad + \frac{1}{1209600n^8} [G^{(7)}(1) - G^{(7)}(0)] - \dots + \text{remainder term}. \end{aligned} \quad (4)$$

The idea becomes clear from this representation of the formula: if it is possible to arrange matters such that the values of $G(q)$ and its low derivatives are either equal and/or zero at the end points then the convergence of the sum over the quadrature points to the exact integral value should become more rapid. Of course in some cases this is true for all derivatives and then all the error will be in the remainder term and nothing is achieved, but in most cases $G(q)$ and its derivatives will not be equal at the end points. Handy and Boys showed how to make the derivatives zero at the end points and specifically considered a Jacobian of the form

$$\frac{dx}{dq} = A_m q^{m-1} (1-q)^{m-1} \quad (5)$$

where A_m is chosen such that the range of q is $[0, 1]$. In their study they considered the first three values of m that removed new odd derivatives and also a transformation which removed all derivatives, $m = \infty$. For the function $\sin(\pi x)$ the fractional errors with $n = 10, 20$ were 823.8×10^{-5} , 205.7×10^{-5} for $m = 1$; 14.7×10^{-5} , 0.93×10^{-5} for $m = 2$; and 1.12×10^{-5} , 0.02×10^{-5} for $m = 3$. These results are typical of the various functions considered in the paper. It is easy to understand why m must not be made too large, because as m is increased the integrand becomes increasingly flat at the ends and more delta-function-like in the middle. Thus $m = \infty$ results were no improvement on $m = 3$. We now discuss the specific use of this approach to each of the spherical polar coordinates, remembering that the Jacobian $r^2 \sin \theta$ is present.

2.1.1. Radial quadrature

The nature of quantum mechanics means that all derivatives of our integrands vanish at $r = \infty$, and the Jacobian factor, r^2 , means the integrand and its first derivative vanish at $r = 0$. We combine the above transformation (5) with a transformation from $[0, \infty]$ to $[0, 1]$, thus

$$r = \alpha q_r^{m_r} (1 - q_r)^{-m_r} \quad (6)$$

$$\frac{dr}{dq_r} = \alpha m_r q_r^{m_r-1} (1 - q_r)^{-m_r-1} \quad (7)$$

where α is a scaling parameter depending upon an effective atomic radius. The use of (6) means that the integrand and its derivatives up to the $(3m - 1)$ th vanish at $q_r = 0$, and all its derivatives vanish at $q_r = 1$.

2.1.2. Theta quadrature

We consider a transformation $\theta \in [0, \pi] \rightarrow q_\theta \in [0, \pi]$ of the form

$$d\theta/dq_\theta = A_{m_\theta} q_\theta^{m_\theta-1} (\pi - q_\theta)^{m_\theta-1} \quad (8)$$

where A_{m_θ} is chosen such that $\theta = \pi$ corresponds to $q_\theta = \pi$. If (8) is used, note that the integrand and its derivatives vanish up to the $(2m - 2)$ th at $q_\theta = 0$ and $q_\theta = \pi$. For specific cases the transformation is

$$m_\theta = 1; \quad \theta = q_\theta \quad (9)$$

$$m_\theta = 2; \quad \theta = q_\theta^2 (3\pi - 2q_\theta) / \pi^2 \quad (10)$$

$$m_\theta = 3; \quad \theta = q_\theta^3 (10\pi^2 - 15\pi q_\theta + 6q_\theta^2) / \pi^4. \quad (11)$$

2.1.3. Phi quadrature

The integrand and all its derivatives will have the same value at $\phi = 0$ and $\phi = 2\pi$, and thus there is no advantage in the above type of transformations. We recommend equally spaced points because it may easily be shown that n such points will exactly integrate $\cos m\phi$ and $\sin m\phi$ for $m = 0, 1, \dots, n - 1$. In this case the transformation is

$$\phi = q_\phi. \quad (12)$$

In our discussion of these quadratures in section 5 we shall therefore be referring to n_r, n_θ, n_ϕ as the number of grid points in each dimension, and also m_r, m_θ which define the transformations in (6) and (8).

2.2. Gauss quadrature schemes

It is hardly necessary to introduce Gauss quadrature [19]. It is probably sufficient to say that it is a quadrature scheme designed to integrate polynomials with specified weight functions exactly. The two such schemes of relevance to this discussion are Gauss-Laguerre and Gauss-Legendre for which

$$\int_0^\infty \exp(-ar) P_l(r) r^2 dr \quad (13)$$

$$\int_0^\pi \sin \theta P_l(\cos \theta) d\theta \quad (14)$$

are integrated exactly for all polynomials P_l of degree $2n - 1$, where n is the number of quadrature points. a is a parameter which in our case may be optimized for each atom.

2.3. Two-dimensional angular quadrature schemes

Becke reports that there is a considerable literature on the numerical evaluation of the two-dimensional integral over a sphere

$$\int_0^\pi \int_0^{2\pi} f(\theta, \phi) \sin \theta \, d\theta \, d\phi. \quad (15)$$

Indeed Maclaren [20] and Lebedev [14] in particular have devised Gauss-like quadrature schemes which exactly integrate the spherical harmonics $Y_{lm}(\theta, \phi)$ for all $-l \leq m \leq l$, $0 \leq l \leq L$, for some L (i.e. all spherical harmonics to degree L). There are $(L + 1)^2$ such functions. Maclaren describes a 72-point quadrature scheme, based on the symmetry of the icosahedron for which $L = 14$. Lebedev describes a 194 point quadrature scheme for which $L = 23$ that is based on the symmetry of the tetrahedron. Although straightforward to implement, the derivation of these schemes is far from trivial (tVD make this observation as well). They are argued to be efficient, with their efficiency (defined by $(L + 1)^2/3N$, where N is the number of points) near unity.

As an alternative, consider the Gauss–Legendre scheme for theta and the simple Gauss scheme for phi, regarded as a product quadrature scheme. To integrate all spherical harmonics of degree $\leq L$, then $(L + 1)/2$ theta points are required and $L + 1$ phi points are required, that is $N = (L + 1)^2/2$. Thus if $L = 14$, $N = 120$ and if $L = 23$, $N = 288$. In other words a factor of about $\frac{3}{2}$ more points are required in this product scheme to integrate exactly the same spherical harmonics. However this product scheme is exceedingly easy to program, the number of points being trivially increased by the change of one parameter. Additionally, as pointed out by tVD, it is far from straightforward to take advantage of all but the simplest symmetry using Maclaren or Lebedev quadrature, whereas it is relatively easy to do with the product scheme. For all these arguments we favour the product scheme, and we consider that the reduction in efficiency by a factor of $\frac{2}{3}$ is not serious.

3. Becke's scheme for multicentre three-dimensional integration

The key to the success of the Becke scheme is the decomposition of the integrand (which we write here simply as $F(\mathbf{r})$) into single centre components through the introduction of weight functions $w_A(\mathbf{r})$ which have a value of near unity near nucleus A and which vanish in a well-behaved manner near any other nucleus. As discussed earlier Boys and Rajagopal [4] recognized the importance of such a procedure. The relevant equations are

$$\sum_A w_A(\mathbf{r}) = 1 \quad (16)$$

$$F(\mathbf{r}) = \sum_A F_A(\mathbf{r}) \quad (17)$$

$$F_A(\mathbf{r}) = w_A(\mathbf{r})F(\mathbf{r}) \quad (18)$$

$$I = \sum_A I_A \quad (19)$$

$$I_A = \int F_A(\mathbf{r}) \, d\mathbf{r}. \quad (20)$$

To determine the weight functions Becke introduced confocal elliptic coordinates (λ, μ, ϕ) for two centres A and B . The key variable is μ defined by

$$\mu_{AB} = \frac{r_A - r_B}{R_{AB}}. \quad (21)$$

μ_{AB} takes the value -1 on A and also along the axis beyond A , and it takes the value $+1$ on B and along the axis beyond B . Becke defined a function $s(\mu_{AB})$ which was $+1$ for $-1 \leq \mu_{AB} \leq 0$ and 0 for $0 < \mu_{AB} \leq +1$. The weight function is then defined by

$$P_A(\mathbf{r}) = \prod_{B \neq A} s(\mu_{AB}) \quad (22)$$

$$w_A(\mathbf{r}) = P_A(\mathbf{r}) / \sum_A P_A(\mathbf{r}). \quad (23)$$

By this scheme the full space has been divided into Voronoi polyhedra surrounding each nucleus A . Becke then smoothed out the discontinuities at $\mu_{AB} = 0$ into continuous mutually overlapping regions. He constructed functions $s(\mu)$ such that $s(-1) = 1$, $s(+1) = 0$, $ds/d\mu = 0$ at $\mu = \pm 1$, to ensure that ' $s(\mu)$ does not have nuclear cusps'. He found it desirable to work with a function for which the first seven derivatives were zero at the ends. This is of course related to our arguments in the previous section about the desirability of having a number of derivatives zero at the end points, and thus it is natural to suggest the following form for s_μ

$$ds/d\mu = A_{m_\mu} (1 - \mu^2)^{m_\mu} \quad (24)$$

with A_{m_μ} being chosen to ensure that $s(\mu)$ has the value $1, 0$ at $\mu = \mp 1$. In the next section we will discuss optimum values for m_μ .

Finally Becke recognized that it is important to have different sizes of regions around each atom, the scheme given so far sharing the space equally between two atoms because $\mu = 0$ corresponds to the mid-point between them. Therefore Becke introduced a change of variable

$$\nu_{AB} = \mu_{AB} + a_{AB}(1 - \mu_{AB}^2) \quad (25)$$

and worked with the function $s(\nu)$ instead of $s(\mu)$, with the boundary at $\nu = 0$. He argued that a good value for a_{AB} is given by

$$a_{AB} = u_{AB}/(u_{AB}^2 - 1) \quad (26)$$

$$u_{AB} = (\chi - 1)/(\chi + 1) \quad (27)$$

$$\chi = R_A/R_B \quad (28)$$

where R_A , R_B are the respective Bragg-Slater radii [21]. We have followed this aspect of Becke's scheme exactly.

Te Velde and Baerends (tVB) also divide space into atomic polyhedra. They then proceed differently, putting an atom centred sphere in the polyhedra and surrounding it by a

number of pyramids. The integration over the spheres uses spherical polar coordinates. Special, more complicated devices are used to integrate over the pyramids. Certainly the Becke smoothing scheme is much easier to use than the tVB scheme.

These two sections have defined the parameters for the quadrature scheme we propose, and it therefore should be possible for any worker to program up our procedure and obtain our values precisely for the integrals (using the actual parameters given in section 5). Before describing how we optimized the parameters introduced in sections 2 and 3, we next give the functions $F(\rho, \nabla\rho)$ whose integrals we evaluate as being typical of those required in density functional theory.

4. Functions whose integrals are evaluated

This section is very short as we simply give the precise forms of the functions whose integrals we evaluate. The most obvious one is the density itself

$$F1 \equiv \rho(\mathbf{r}). \quad (29)$$

In all our examples we use the converged SCF density, and therefore the integral of $F1$ is the number of electrons. All electrons are active and we have no core assumed. The second function defines the local density approximation (LDA) for the exchange energy [22]

$$F2 \equiv -\frac{3}{4}(3/\pi)^{1/3}\rho^{4/3}. \quad (30)$$

The next function gives the non-local gradient correction to the exchange energy, we use the form due to Becke [23]:

$$F3 \equiv -\beta \sum_{\sigma} \rho_{\sigma}^{4/3} \frac{x_{\sigma}^2}{1 + 6\beta x_{\sigma} \sinh^{-1} x_{\sigma}} \quad (31)$$

where

$$x_{\sigma} = \frac{|\nabla\rho_{\sigma}|}{\rho_{\sigma}^{4/3}} \quad (32)$$

and σ denotes α or β spins. We use Becke's recommended value for β of 0.0042. Now we turn to correlation functionals. We chose to use the non-local correlation correction due to Perdew [24]

$$F4 \equiv \rho\epsilon_c(\rho_{\alpha}, \rho_{\beta}) + d^{-1} \exp(-\Phi)C(\rho)|\nabla\rho|^2/\rho^{4/3} \quad (33)$$

where

$$\Phi = 1.745 \times 0.11 \times C(\infty)|\nabla\rho|/[C(\rho)\rho^{7/6}] \quad (34)$$

$$d = 2^{1/3} \left[\left(\frac{1+\zeta}{2} \right)^{5/3} + \left(\frac{1-\zeta}{2} \right)^{5/3} \right]^{1/2} \quad (35)$$

$$\zeta = (\rho_{\alpha} - \rho_{\beta})/\rho \quad (36)$$

$$\rho = \rho_{\alpha} + \rho_{\beta} \quad (37)$$

$$C(\rho) = 0.001667 + \frac{(0.002568 + \alpha r_s + \beta r_s^2)}{(1 + \gamma r_s + \delta r_s^2 + 10^4 \beta r_s^3)} \quad (38)$$

$$\rho = (4\pi r_s^3/3)^{-1}. \quad (39)$$

Values are $\alpha = 0.023266$, $\beta = 7.389 \times 10^{-6}$, $\gamma = 8.723$, $\delta = 0.472$. Finally we have chosen to use the correlation energy per particle of the uniform electron gas [25] $\epsilon_c(\rho_\alpha, \rho_\beta)$ of Ceperley and Alder [26, 27]. This is given by (a) for high densities ($r_s < 1$)

$$\epsilon_c = A \ln r_s + B + Cr_s \ln r_s + Dr_s \quad (40)$$

where [26, 28] $A = (0.0311, 0.01555)$ and $B = (-0.048, -0.0269)$, the two arguments corresponding to the unpolarized and polarized cases respectively. Similarly $C = (0.0020, 0.0007)$, $D = (-0.0116, -0.0048)$. For (b) low density ($r_s \geq 1$)

$$\epsilon_c = \gamma/[1 + \beta_1(r_s)^{1/2} + \beta_2 r_s] \quad (41)$$

where the constants are $\gamma = (-0.1423, -0.0843)$, $\beta_1 = (1.0529, 1.3981)$, $\beta_2 = (0.3334, 0.2611)$.

This completes our descriptions of the functions used in the next section.

5. Accuracy and the integration parameters

To determine an optimum number of quadrature points and the other parameters introduced in sections 2 and 3, we can only turn to specific examples.

5.1. Radial quadrature parameters m_r , n_r

In tables 1 and 2, we report investigations on neon using a TZ basis set derived from the Dunning 5s4p contraction [29] of a Huzinaga 10s6p primitive set [30]. We have not found that our conclusions vary significantly with basis set. For neon we are only investigating the radial quadrature. We first investigate the Euler–Maclaurin scheme given by (6) and (7). Our test functions were $F1$, (28) and $F2$ (29). We have found that a good value for α is the Bragg radius [21], although for neon a value is not given in this reference, we can extrapolate to give 0.45 Å. This contrasts with Becke [12] who suggests one half the Bragg radius for a related parameter. The larger value is used here because as the number of quadrature points n_r increases, the furthest points contribute negligibly to the integrand and we are able to ignore them. This is done by making note of the contribution of the integrand density from each radial shell as the shells becomes larger, when this contribution falls below 10^{-12} atomic units all points further out from the nucleus are rejected. The test is only applied to points that are more than 5 Å from the nucleus. Thus with $m_r = 2$, $n_r = 40$, (m_r is the parameter in (6)) only 30 points are used in practice. In table 1 we examine the dependence of the accuracy on the parameter m_r , for $m_r = 1, 2$ and 3. The table shows that best results are obtained with $m_r = 2$ for all values of n_r . We use this value of m_r throughout.

Table 2 examines in greater detail how the accuracy depends on the number of quadrature points. The table shows that acceptable results (10^{-15} accuracy) are achieved with $n_r > 30$. Also in table 2 we compare our calculations with those using Gauss–Laguerre points, with a partially optimized exponent of 12.0. As far as this test is concerned, the Euler–Maclaurin scheme appears preferable. One reason for this is that here we are integrating two radial shells; it appears that the Euler–Maclaurin scheme can cope with multiple shells, it is not surprising that Gauss–Laguerre cannot. Indeed tVD also recognize this problem, and recommend that the radial integration is split into several intervals with different quadratures in each

Table 1. Investigation of convergence of radial quadrature for different values of m_r , equation (6), in the Euler–Maclaurin scheme for neon using a 5s4p basis. Errors in the integrals of $F1(=10)$ and $F2(=-11.038076569)$ are tabulated for different numbers of significant radial points, n_r (with the total number of radial points in parentheses).

n_r	$m_r = 1$			$m_r = 2$			$m_r = 3$		
	$\Delta F1$	$\Delta F2$	n_r	$\Delta F1$	$\Delta F2$	n_r	$\Delta F1$	$\Delta F2$	n_r
13 (14)	1×10^{-2}	1×10^{-1}	11 (14)	7×10^{-3}	2×10^{-3}	10 (14)	1×10^{-2}	2×10^{-2}	
17 (18)	6×10^{-3}	5×10^{-2}	14 (18)	6×10^{-3}	3×10^{-3}	13 (18)	5×10^{-3}	2×10^{-2}	
20 (22)	2×10^{-3}	2×10^{-2}	17 (22)	9×10^{-7}	3×10^{-4}	15 (22)	1×10^{-2}	2×10^{-3}	
24 (26)	1×10^{-3}	4×10^{-3}	20 (26)	6×10^{-5}	1×10^{-5}	18 (26)	9×10^{-3}	4×10^{-3}	
29 (32)	6×10^{-4}	5×10^{-3}	24 (32)	1×10^{-5}	3×10^{-7}	22 (32)	2×10^{-4}	2×10^{-4}	
36 (40)	2×10^{-4}	2×10^{-3}	30 (40)	3×10^{-8}	4×10^{-7}	27 (40)	6×10^{-5}	1×10^{-5}	
43 (48)	6×10^{-5}	6×10^{-4}	36 (48)	8×10^{-9}	9×10^{-8}	33 (48)	4×10^{-6}	4×10^{-6}	

Table 2. Comparison of the convergence of the radial quadrature for Gauss–Laguerre and Euler–Maclaurin schemes for neon. Other details as for table 1.

Euler–Maclaurin			Gauss–Laguerre		
n_r	$\Delta F1$	$\Delta F2$	n_r	$\Delta F1$	$\Delta F2$
12 (15)	1×10^{-2}	7×10^{-4}	12 (12)	4×10^{-3}	2×10^{-3}
14 (18)	6×10^{-3}	3×10^{-3}	14 (14)	2×10^{-3}	4×10^{-3}
16 (21)	8×10^{-4}	7×10^{-4}	16 (16)	4×10^{-5}	5×10^{-5}
20 (27)	6×10^{-5}	1×10^{-5}	20 (20)	8×10^{-4}	4×10^{-3}
24 (32)	1×10^{-5}	3×10^{-7}	24 (24)	6×10^{-4}	2×10^{-3}
30 (40)	3×10^{-8}	4×10^{-7}	30 (30)	1×10^{-4}	4×10^{-4}
37 (50)	7×10^{-10}	1×10^{-8}	37 (37)	6×10^{-5}	3×10^{-4}

interval. Our investigations on helium showed that Gauss–Laguerre was preferable for that atom, which is no surprise, but investigations on H_2 showed that Euler–Maclaurin was only marginally more costly than Gauss–Laguerre. We have also tested the use of Gauss–Laguerre for the hydrogens only, in the water molecule and have found that there is no advantage in comparison to the Euler–Maclaurin method. For these reasons we decided to use the Euler–Maclaurin scheme as described for radial quadrature for all atoms.

5.2. The smoothing parameter m_μ and the theta quadrature parameters m_θ , n_θ

We shall demonstrate these investigations with N_2 at an internuclear distance of 1.3 Å. The Bragg radius is 0.65 Å [21]. We used the 6-311G* basis set [31], we have verified that our conclusions are independent of basis set. Because we are investigating a homonuclear diatomic, a_{AB} , (25) is zero and therefore the smoothing function is given by $s(\mu)$, (equation (23)). In table 3 we investigate the accuracy dependence on m_μ , using a range of values for the number of radial and angular quadrature

Table 3. Investigation of influence of smoothing parameter m_μ , equation (23), on the errors in the integrals of $F1(= 14)$ and $F2(= -11.6264121833)$ for N_2 using a 6-311G* basis, at $R = 1.3$ Å. The results for different numbers of radial points, n_r , and θ points, n_θ , are tabulated.

m_μ	$n_r = 16$ (20), $n_\theta = 15$		$n_r = 25$ (30), $n_\theta = 20$		$n_r = 37$ (48), $n_\theta = 32$	
	$\Delta F1$	$\Delta F2$	$\Delta F1$	$\Delta F2$	$\Delta F1$	$\Delta F2$
0	1.3	3.1	1.4	3.2	0.5	1.6
1	1×10^{-1}	2×10^{-1}	1×10^{-1}	2×10^{-1}	5×10^{-2}	2×10^{-1}
2	2×10^{-2}	4×10^{-2}	1×10^{-2}	2×10^{-2}	3×10^{-3}	1×10^{-2}
3	4×10^{-3}	6×10^{-3}	7×10^{-4}	3×10^{-3}	8×10^{-4}	3×10^{-4}
4	2×10^{-4}	2×10^{-4}	4×10^{-4}	7×10^{-5}	5×10^{-4}	6×10^{-4}
5	1×10^{-4}	7×10^{-4}	3×10^{-4}	2×10^{-4}	2×10^{-4}	2×10^{-4}
8	3×10^{-3}	7×10^{-4}	2×10^{-4}	1×10^{-4}	5×10^{-6}	5×10^{-7}
10	5×10^{-3}	2×10^{-3}	2×10^{-4}	1×10^{-4}	4×10^{-6}	3×10^{-6}
13	5×10^{-3}	2×10^{-3}	2×10^{-4}	1×10^{-4}	1×10^{-6}	7×10^{-7}
14	8×10^{-2}	1×10^{-1}	6×10^{-2}	1×10^{-2}	3×10^{-2}	8×10^{-2}
∞	1×10^{-1}	7×10^{-2}	3×10^{-1}	1×10^{-1}	2×10^{-2}	2×10^{-2}

Table 4. Comparison of the theta convergence for Gauss–Legendre and Euler–Maclaurin (with three different values of m_θ , equation (8)) quadrature for N_2 . $n_r = 128$ and other details are as in table 3.

n_θ	Euler–Maclaurin quadrature					
	$m_\theta = 1$		$m_\theta = 2$		$m_\theta = 3$	
	$\Delta F1$	$\Delta F2$	$\Delta F1$	$\Delta F2$	$\Delta F1$	$\Delta F2$
8	5×10^{-4}	2×10^{-3}	5×10^{-3}	3×10^{-3}	3×10^{-4}	1×10^{-2}
12	1×10^{-3}	1×10^{-3}	2×10^{-3}	3×10^{-4}	4×10^{-3}	2×10^{-3}
16	3×10^{-4}	3×10^{-4}	1×10^{-4}	1×10^{-6}	3×10^{-4}	2×10^{-4}
20	1×10^{-4}	1×10^{-4}	5×10^{-6}	1×10^{-6}	7×10^{-6}	2×10^{-5}
24	6×10^{-5}	6×10^{-5}	1×10^{-6}	1×10^{-6}	2×10^{-6}	3×10^{-6}
32	2×10^{-5}	2×10^{-5}	2×10^{-7}	1×10^{-7}	5×10^{-8}	3×10^{-8}
40	8×10^{-6}	7×10^{-6}	4×10^{-8}	4×10^{-8}	8×10^{-10}	4×10^{-10}
64	1×10^{-6}	1×10^{-6}	2×10^{-9}	2×10^{-9}	0	0

Gauss–Legendre quadrature		
n_θ	$\Delta F1$	$\Delta F2$
8	4×10^{-4}	7×10^{-5}
10	5×10^{-5}	3×10^{-6}
12	2×10^{-6}	6×10^{-6}
16	6×10^{-7}	3×10^{-7}
20	4×10^{-8}	6×10^{-8}
24	2×10^{-8}	2×10^{-8}
32	1×10^{-9}	1×10^{-9}
48	0	0

points n_r, n_θ (we are using the Euler–Maclaurin theta integration scheme with $m_\theta = 3$, equation (11)). The results clearly show that best results are achieved for $4 \leq m_\mu \leq 12$, with larger values of m_μ being appropriate for larger numbers of quadrature points. Note that the results underline the importance (stressed by Becke) of having a smoothing function, as demonstrated by the poor results for both $m_\mu = 0$, the straight line case, and $m_\mu = \infty$, the step function case. Because of the relative insensitivity of the results from hereon we use $m_\mu = 10$.

We next investigate and compare the accuracy of our theta quadrature schemes. Thus we examine the dependence on m_θ (8) for the Euler–Maclaurin scheme, and the number of points n_θ for both schemes. To do this we use a large number of radial points ($n_r = 128$). Results are given in table 4 for $m_\theta = 1, 2$ and 3 and for the Gauss–Legendre quadrature (14). These results demonstrate that best accuracy of the Euler–Maclaurin scheme is obtained with $m_\theta = 2$, it is seen that the $F2$ error is significantly greater for $m_\theta = 1$. Table 4 is also sufficient to show that good accuracy (10^{-6}) will be achieved with approximately $n_\theta = 20$. However more importantly, it is clear from table 4 that the accuracy achievable with Gauss–Legendre quadrature is greater, as with $n_\theta = 20$ an accuracy of 10^{-8} is obtained. We therefore decided to use Gauss–Legendre quadrature for the theta integration.

The next question we address is whether it is possible to reduce crowding of quadrature points near nuclei. In other words when r is small it should be possible

Table 5. Investigation of the accuracy of angular integration with H_2O as an example ($0.9584 \text{ \AA}$, 104.45° , $6\text{-}31\text{G}^*$), with 128 radial points on each atom. % gives the percentage of angular points actually used because of the application of the angular crowding test in equation (42) with $K_\theta = 5$. The errors in the integration of $F1$ ($= 10$) and $F2$ ($= -8.11530440048$) are tabulated.

n_θ	n_ϕ	%	$\Delta F1$	$\Delta F2$
8	16	71	8×10^{-4}	1×10^{-4}
10	20	70	3×10^{-4}	4×10^{-5}
12	24	69	9×10^{-5}	2×10^{-5}
16	32	68	8×10^{-6}	1×10^{-6}
20	40	68	9×10^{-7}	1×10^{-7}
24	48	68	9×10^{-8}	1×10^{-8}
32	64	68	1×10^{-9}	2×10^{-10}
48	96	69	1×10^{-10}	—

to reduce the number of theta (and phi, see next section) points. The formula we have used is

$$n_\theta = \min\left(\frac{K_\theta r}{r_{\text{Bragg}}} n_{\theta_{\text{max}}}, n_{\theta_{\text{max}}}\right). \quad (42)$$

Table 5 illustrates the use of this formula but discussion of the table will be put off until integration over the phi direction has been covered.

5.3. The phi quadrature parameter n_ϕ

For the phi integration, we use equally spaced points in phi space, and following the discussion in section 2.3, we always use twice as many phi points as theta points for systems where there is no reduction in quadrature points through symmetry. We shall demonstrate these investigations on H_2O at the geometry ($0.9584 \text{ \AA}$, 104.45°) with a $6\text{-}31\text{G}^*$ basis set [31]. We use $n_r = 128$ and this should give a very accurate integration over the radial direction as seen from our results. Table 5 shows that good accuracy is obtained with relatively few angular points as has been discovered by many other authors. The theta crowding test of (42) is applied with $K_\theta = 5$ and the use of this test also reduces the number of phi points, because we always use twice as many phi as theta points. This test reduces the overall number of points used by 30% with no loss in accuracy to twelve decimal places, and is therefore to be recommended for calculations of this type. It is worth pointing out that Andzelm and Wimmer [15] employ a test based on similar principles which is more sophisticated but requires a preliminary assessment of accuracy of the quadrature on a trial density based on a sum of atomic contributions.

Thus far, we have neglected to mention the direction of the polar axis which is necessary to define the position of the angular quadrature points. We have defined it to be parallel to the symmetry axis, for all atoms. Our accuracy should not of course depend upon the direction of this axis, and we have verified this to some extent. This leads us on to a discussion of the effects of symmetry.

6. The implications of symmetry

It is well known that symmetry can be used to reduce the number of quadrature

points in numerical integration schemes [10, 11]. For most molecules the density is totally symmetric and consequently all the functions of the density that we have considered are also totally symmetric. This means that the contribution to the quadrature from a point P will be the same as that from the set of points $P_i = R_i P$ where R_i are the group operations of the point group to which the molecule belongs. Therefore in the limit of a very large number of quadrature points, the number that are necessary reduces by a factor N_g , the order of the group, and this restricted summation will equal the full summation when multiplied by the symmetry factor N_g . Of course, this requires that the grid be chosen so that it has the symmetry of the point group which may not always be convenient. In our quadrature scheme, the splitting of the quadrature into θ and ϕ components prevents the choice of a fully symmetric grid in all cases. However we have taken advantage of some symmetry by the following simple two step procedure.

The quadrature program contains a loop over the atoms. For each atom the operation of the group elements on its coordinates is performed and if the coordinates are transformed into the position of another atom which occurs later in the loop over atoms, the current atom is missed out. For the last symmetry equivalent atom in the list the number of other atoms it transforms into is noted, and this is a symmetry factor which multiplies the weighting for the quadrature points.

For each atom that survives this test a random point (r_0, θ_0, ϕ_0) close to the nucleus is chosen; it is important that this point does not transform into itself under any of the group operations (except the identity). If under the group operations it transforms to a point (r_0, θ_i, ϕ_0) then this indicates a plane of symmetry or a two-fold axis that can be exploited. (These are the only possibilities for molecules.) The symmetry factor is multiplied by two and if the number of θ points is chosen to be even, only positive points in the Gauss-Legendre quadrature need be used. If under group operations the starting point transforms to a set of $n - 1$ distinct points of the form (r_0, θ_0, ϕ_i) then this indicates an n -fold symmetry axis (or if $n = 2$ then it could be a reflection plane). The symmetry factor is multiplied by n and a restriction in the number of ϕ points is possible. We achieve this by making the number of ϕ points divisible by n , with $\phi = 0$ along the perpendicular from the atom to the symmetry axis. We go through each ϕ value in turn keeping a record of only one of the symmetry equivalent ϕ points and treating ϕ values that transform into themselves in a special but obvious way. Having made a note of these restrictions on the θ and ϕ points, the loops over the radial and angular points can now be completed as normal.

The results for grids A and B in table 8 give an indication of the sort of saving in the number of points that is achieved by this simple approach. The symmetry axis for these molecules is always chosen to lie along the z -axis, in reality this is not a significant restriction. Full symmetry reduction is obtained in all cases except CH_4 and SF_6 . For methane the symmetry factor is 4 on the carbon atom and 8 on the hydrogens as compared with a possible 24 on all atoms; for SF_6 the symmetry factor is 16 on the sulphur and 48 on the fluorines as compared with a possible 48 for all atoms. For all the molecules the use of symmetry results in a substantial reduction in the number of quadrature points. It is noted that this straightforward implementation of symmetry is made possible by the simple product scheme that we employ for angular integration. Schemes based on polyhedra (that is Maclaren [20] or Lebedev [14] quadrature) are much less attractive from this point of view since one is constrained to consider only point groups that are subgroups of the polyhedra.

7. Applications to molecules

After taking into account the studies of the previous sections we favour the following quadrature scheme: for the radial quadrature, Euler–Maclaurin is used with m_r of (6) being 2; for theta quadrature we use Gauss–Legendre and employ (42) to reduce crowding near the nucleus with $K_\theta = 5$; for the phi direction we use twice as many equally spaced points. We follow Becke’s scheme for the partitioning of space employing $m_\mu = 10$ in (23) for the smoothing parameter. We use symmetry as in the previous section unless otherwise stated.

Having chosen our scheme by a consideration of detailed studies on H_2O , N_2 and neon, it is important to test it on a more general selection of molecules. In the end, we hope this will allow a fair comparison of our scheme with the approaches of other workers and thus arrive at a good general method for numerical quadrature in molecular applications of density functional theory. In table 6 we give detail on the different grids used in our studies. Grids A and B differ only in their use of symmetry, and grid D was chosen to have roughly ten times as many points as B with grid C being intermediate. It should be remembered when looking at the number of radial points in table 6 that the number that will be actually used in calculations will be about 20–25% lower than this. Table 7 gives accurate values for the integration of the four functions described in section 4 for ten different molecules. These results were obtained by using a larger grid than those described in table 6 and should be at least an order of magnitude more accurate than those obtained with grid D although they are not all accurate to the number of figures quoted in the table. The densities are from a 6-31G* SCF calculation and the

Table 6. Description of grid types used in tables 7 and 8. n_{r0} , n_{r1} and n_{r2} are the number of radial points on hydrogen, first row atom and second row atoms respectively.

Grid	Symmetry	n_{r0}	n_{r1}	n_{r2}	n_θ	n_ϕ
A	OFF	20	25	30	8	16
B	ON	20	25	30	8	16
C	ON	31	36	41	12	24
D	ON	46	54	60	16	32

Table 7. Accurate values for the integrals of $F1$, $F2$, $F3$ and $F4$ for various molecules using the SCF density with a 6-31G* basis at their SCF optimized geometries.

Molecule	$F1$	$F2$	$F3$	$F4$
H_2O	10	−8.125929484	−0.868507026	−0.362921744
H_2S	18	−23.66992536	−2.033207369	−0.753212445
H_2O_2	18	−15.56748374	−1.651653983	−0.670708591
H_2CO	16	−12.74408706	−1.407520364	−0.570041040
NH_3	10	−6.926858287	−0.768584690	−0.347276232
CH_4	10	−5.890199354	−0.685350830	−0.327078322
C_2H_4	16	−10.54454594	−1.214621167	−0.541819718
C_2H_6	18	−11.20566235	−0.609626784	−0.609626784
C_6H_6	42	−29.06672389	−3.464404395	−1.409421266
SF_6	70	−78.41102803	−7.479085436	−2.868734817

Table 8. Comparison of the errors in the integrals of $F1$, $F2$, $F3$ and $F4$ relative to the accurate values in table 7 for the grids described in table 6. The number of grid points for the whole molecule is also given.

Molecule	Grid	Points	$\Delta F1$	$\Delta F2$	$\Delta F3$	$\Delta F4$
H ₂ O	A	4696	2×10^{-4}	1×10^{-4}	8×10^{-5}	2×10^{-4}
H ₂ O	B	1394	2×10^{-4}	1×10^{-4}	8×10^{-5}	2×10^{-4}
H ₂ O	C	4380	5×10^{-5}	4×10^{-5}	6×10^{-7}	5×10^{-6}
H ₂ O	D	10906	2×10^{-6}	3×10^{-7}	1×10^{-7}	8×10^{-7}
H ₂ S	A	5216	2×10^{-3}	5×10^{-4}	6×10^{-4}	2×10^{-4}
H ₂ S	B	1550	2×10^{-3}	4×10^{-4}	6×10^{-4}	2×10^{-4}
H ₂ S	C	4648	2×10^{-4}	2×10^{-5}	5×10^{-5}	5×10^{-5}
H ₂ S	D	11154	1×10^{-5}	7×10^{-5}	7×10^{-7}	2×10^{-7}
H ₂ O ₂	A	6400	2×10^{-4}	3×10^{-4}	2×10^{-4}	2×10^{-4}
H ₂ O ₂	B	3200	2×10^{-4}	3×10^{-4}	2×10^{-4}	2×10^{-4}
H ₂ O ₂	C	10504	7×10^{-5}	3×10^{-5}	6×10^{-7}	2×10^{-5}
H ₂ O ₂	D	27320	3×10^{-7}	2×10^{-7}	6×10^{-8}	2×10^{-6}
H ₂ CO	A	6656	4×10^{-4}	7×10^{-4}	7×10^{-6}	2×10^{-4}
H ₂ CO	B	2010	4×10^{-4}	7×10^{-4}	7×10^{-6}	2×10^{-4}
H ₂ CO	C	5976	1×10^{-4}	6×10^{-5}	3×10^{-6}	2×10^{-5}
H ₂ CO	D	14930	5×10^{-6}	1×10^{-6}	1×10^{-6}	8×10^{-7}
NH ₃	A	6192	1×10^{-4}	5×10^{-5}	3×10^{-5}	3×10^{-4}
NH ₃	B	1364	5×10^{-4}	1×10^{-4}	5×10^{-5}	3×10^{-4}
NH ₃	C	3948	3×10^{-4}	1×10^{-4}	8×10^{-6}	2×10^{-5}
NH ₃	D	10054	1×10^{-6}	2×10^{-6}	2×10^{-7}	2×10^{-6}
CH ₄	A	8200	3×10^{-3}	9×10^{-4}	4×10^{-5}	1×10^{-4}
CH ₄	B	1452	3×10^{-3}	9×10^{-4}	4×10^{-5}	1×10^{-4}
CH ₄	C	4354	3×10^{-4}	1×10^{-4}	1×10^{-5}	1×10^{-5}
CH ₄	D	10862	4×10^{-6}	8×10^{-7}	5×10^{-8}	1×10^{-6}
C ₂ H ₄	A	9904	1×10^{-3}	7×10^{-4}	2×10^{-4}	9×10^{-5}
C ₂ H ₄	B	1407	1×10^{-3}	7×10^{-4}	2×10^{-4}	9×10^{-5}
C ₂ H ₄	C	4257	3×10^{-4}	2×10^{-4}	3×10^{-5}	3×10^{-5}
C ₂ H ₄	D	10672	8×10^{-7}	6×10^{-6}	5×10^{-7}	6×10^{-7}
C ₂ H ₆	A	13152	1×10^{-2}	3×10^{-3}	6×10^{-4}	1×10^{-4}
C ₂ H ₆	B	1260	1×10^{-2}	3×10^{-3}	7×10^{-4}	2×10^{-4}
C ₂ H ₆	C	3930	1×10^{-3}	3×10^{-4}	5×10^{-5}	2×10^{-5}
C ₂ H ₆	D	10010	1×10^{-4}	2×10^{-5}	1×10^{-5}	2×10^{-6}
C ₆ H ₆	A	19968	3×10^{-2}	7×10^{-3}	6×10^{-3}	1×10^{-3}
C ₆ H ₆	B	946	4×10^{-2}	7×10^{-3}	5×10^{-3}	5×10^{-4}
C ₆ H ₆	C	2943	4×10^{-3}	1×10^{-3}	2×10^{-4}	6×10^{-4}
C ₆ H ₆	D	7431	3×10^{-4}	3×10^{-6}	7×10^{-5}	4×10^{-5}
SF ₆	A	12064	9×10^{-3}	3×10^{-3}	6×10^{-5}	6×10^{-4}
SF ₆	B	545	1×10^{-2}	3×10^{-3}	1×10^{-4}	5×10^{-4}
SF ₆	C	1460	7×10^{-3}	4×10^{-3}	6×10^{-4}	1×10^{-4}
SF ₆	D	3402	1×10^{-4}	1×10^{-4}	9×10^{-5}	1×10^{-4}

geometries are all 6-31G* optimized structures at the SCF level which were mainly obtained from reference [31].

Table 8 contains the absolute errors in the integration for the different grids with the ten molecules of table 7. Most trends are not surprising. The accuracy increases

as the number of points is increased. Absolute errors are larger for larger molecules which is expected since it is the relative error that is the size extensive property. However in chemistry we are usually interested in energy differences so it is the absolute error that is the more important quantity. The error in the integration of the density, $F1$, appears to an excellent guide to the errors in the other integrals, with the error in the density being greater for nearly all cases.

We are actively using this quadrature scheme in DFT studies. As a demonstration we have been concerned with the determination of the 20 unique harmonic frequencies of benzene [32]. A large basis set is required, and we used 288 basis functions. We calculated the frequencies by numerical differentiation using central differences of DFT energy functionals established with the SCF density. This required energies to be evaluated to an accuracy of 10^{-10} hartree if frequencies were to be reliable to 1 cm^{-1} . Based on experience with diatomics, we had hoped that less accurate quadrature could be used because of a cancellation of errors in the finite difference approach, but this proved not to be the case. We found that our quadrature scheme could achieve the required accuracy, giving the integrated density as 42.00000000002. We used 164 radial points, 64 theta points and 128 phi points. This may seem a large number of points, but the time to calculate the DFT energy was one third of the time to calculate the Møller–Plesset second-order (MP2) energy and the results are at least as good if not better. Another approach, which is in some ways more attractive, is to establish frequencies by finite differences of analytic gradients [33] although this is only feasible at the moment for densities that are self consistent with their energy expression. Such an approach puts much less strain on the quadrature scheme and sufficiently accurate frequencies can be calculated with fewer points. For this reason we are actively developing Kohn–Sham codes, and the establishment of an efficient and flexible quadrature method is a stepping stone in this direction. We note that this accuracy was easily obtainable because of the flexibility of our scheme and it would not have been possible with any existing two-dimensional angular scheme.

We hope that this paper contributes a little to the practical implementation of density functional theory. We believe that the problem of evaluation of the integrals which arise in DFT theory is not a difficult one, if numerical quadrature is used. Our calculations show that the desirable accuracy is easily achieved. In particular we have demonstrated that our scheme is capable of producing any desired accuracy, and we also believe that our radial quadrature is the best yet available. Finally we add that all of our ideas can be extended to the evaluation of the integrals which arise in Kohn–Sham theory, and therefore also the evaluation of the gradient of the DFT energy.

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